

# Esau Salvador Martinez Lopez

AI Engineer

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## Profile Summary

Software Engineer with 5 years building production systems for government and legal tech, specializing in AI. Currently leading R&D at the State Prosecutor's Office of Guanajuato, Mexico, where forensic tools I developed are being evaluated by multiple jurisdictions. Previously built AI-powered workflows serving 1,000+ customers at a US legal tech company.

## Professional Experience

**AI Engineer** Mar 2026 – Present – On-site  
State Prosecutor's Office of Guanajuato, Mexico

- Lead R&D initiatives developing AI-powered forensic tools. Manage an on-premise cluster of 8 NVIDIA H100 GPUs with vLLM for model deployment, KV-cache optimization, and performance tuning.
- Now heading the technical direction of the forensic AI portfolio, overseeing ongoing projects and adoption efforts.
- Developed a **tattoo-based person search system** using image embeddings for forensic identification in SEMEFO (forensic services).
- Co-developed **“Rostros que no se olvidan”**, a forensic aging system using computer vision that achieves 94% accuracy with 248+ progressions. In evaluation by 4 additional state prosecutor offices.
- Developed a **phone records analysis platform** with interactive statistics and maps for investigators.

**Backend Software Engineer** Dec 2021 – Feb 2026 – Remote  
Justia — Legal Technology (US)

- Designed backend services in TypeScript (Node.js) and AWS Lambda for large-scale legal automation, serving 1,000+ customers.
- Built a **semantic search system** using vector embeddings and AI vector databases, enabling relevance-based document retrieval.
- Developed an **AI Writer pipeline** using LangChain and LangGraph, orchestrating multiple LLMs (OpenAI, Gemini) for structured content creation.
- Built an **AI-driven legal chatbot agent** using LangChain and LangGraph for structured question creation and legal domain classification.
- Designed event-driven pipelines using Redis-backed queues and AWS Step Functions for multi-stage workflow orchestration.

## Technical Skills

**AI & LLMs:** OpenAI, Gemini, LangChain, LangGraph, prompt engineering, multi-LLM orchestration, AI agents, machine learning, computer vision

**ML Infrastructure:** NVIDIA H100 GPU clusters, model deployment, PyTorch, HuggingFace, MLOps

**Computer Vision:** facial aging models, image embeddings, semantic similarity search, tattoo recognition, OpenCV

**Cloud & Infrastructure:** AWS (Lambda, Step Functions, EC2, S3), Docker, Kubernetes, CI/CD, microservices

**Backend & Frameworks:** Node.js, Express.js, AdonisJS, REST APIs

**Data Stores:** MySQL, Redis (queues, caching), Amazon Redshift (columnar analytics)

**Programming Languages:** TypeScript, JavaScript (ES6+), Python

## Education

**Universidad Tecnológica de León** 2018 – 2023  
B.S. in Software Engineering

**Université Grenoble Alpes** 2020 – 2021  
International Exchange Program — Software Engineering

## Languages

**Spanish:** Native   **English:** B2   **French:** B2